

MODULE 1 · TEXTBOOK 1 · CHAPTERS 1 & 2 · 9 HOURS

Computer Graphics Hardware & Software

Display devices and systems, plus the software foundations of graphics: coordinate representations, graphics functions, and an introduction to OpenGL with the GLUT toolkit.

Video Display Devices

Raster & Random Scan

CRT · LCD · Flat Panel

OpenGL & GLUT

Module Roadmap

Two halves: the hardware that shows pixels, and the software that commands them

Part A · Graphics Hardware

- Video display devices
- Raster scan systems
- Random scan (vector) displays
- Colour CRT monitors
- Flat panel displays and LCDs
- Graphics workstations and viewing systems

Part B · Graphics Software

- Coordinate representations
- Graphics functions
- Introduction to OpenGL
- Related libraries and GLUT
- A complete OpenGL program
- Error handling in OpenGL

What is Computer Graphics?

Creating, storing and manipulating pictures with a computer

A picture on screen is a grid of tiny coloured squares called **pixels** (picture elements). The colour of every pixel is held in memory called the **frame buffer**, and the display hardware reads this memory many times per second to keep the image visible.



Interactive

Games, VR and CAD, where the user changes the image in real time.



Passive

Rendered films and pre-computed animation that the user only views.



Visualization

Medical imaging, scientific and data visualization.



Design

CAD and CAM for engineering and architecture.

A picture is a grid of pixels:



Zoom out and the pixels blend into a smooth image

Resolution is the number of pixels; more pixels means a sharper picture.

Video Display Devices

The monitor is the primary output device of a graphics system

Most traditional monitors use the **Cathode Ray Tube (CRT)** design, while modern displays use flat panel technologies. In all of them the picture must be **refreshed** repeatedly because the glowing phosphor or the pixel state fades.

60 Hz+

Typical refresh rate to avoid flicker

Persistence

How long phosphor keeps glowing

Resolution

Number of addressable pixels

Aspect ratio

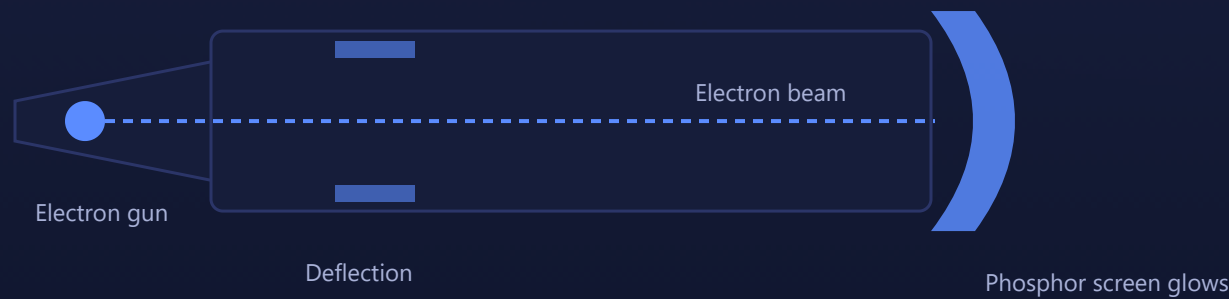
Horizontal to vertical pixel ratio

REFRESH RATE

The number of times per second the whole picture is redrawn. If it is too low the image appears to flicker, so 60 Hz or higher is used.

The Cathode Ray Tube (CRT)

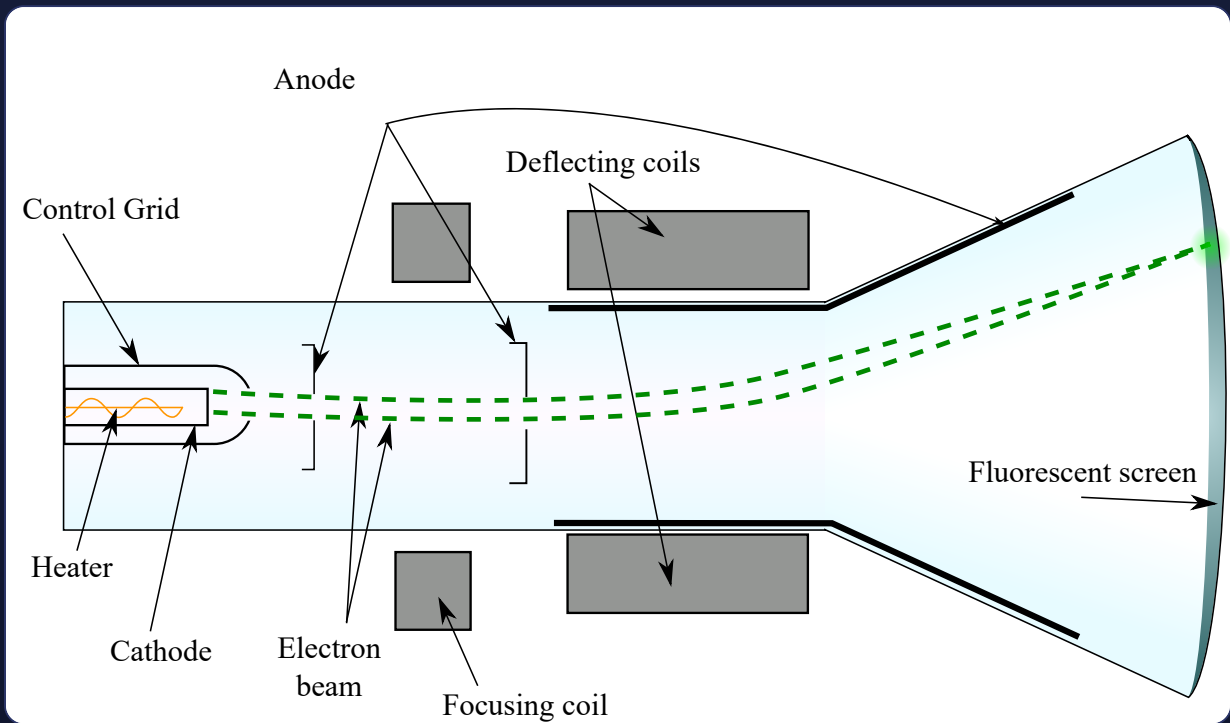
A beam of electrons draws light on a phosphor coated screen



Heated cathode emits electrons, which are focused, deflected and strike the phosphor.

- 1 A **heated cathode** (the electron gun) emits a stream of electrons.
- 2 **Focusing** systems squeeze the beam into a tiny spot.
- 3 **Deflection** (magnetic or electric) aims the beam at any screen position.
- 4 The beam hits the **phosphor**, which glows. The **control grid** voltage sets the brightness.

REFERENCE IMAGE



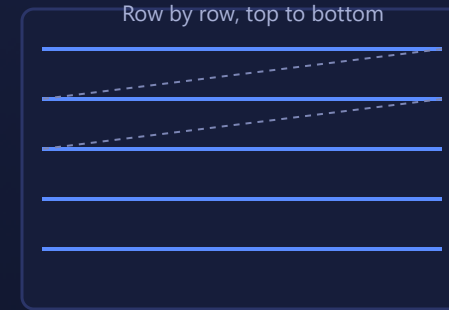
Cathode ray tube diagram. Source: Interior / Theresa Knott, Wikimedia Commons, CC BY-SA 3.0.

Raster Scan Systems

The most common technique: the screen is a matrix of pixels

The electron beam sweeps the screen **one horizontal row at a time**, from top to bottom. Each row is a **scan line**. The picture is stored in the **frame buffer**, where every pixel has a stored intensity or colour.

- **Bitmap**: one bit per pixel, black and white.
- **Pixmap**: multiple bits per pixel for colour.
- **Horizontal retrace**: beam jumps back to the left after each line.
- **Vertical retrace**: beam jumps to the top after each full frame.
- **Interlacing**: refresh odd lines, then even lines, to reduce flicker.



Solid lines: beam drawing. Dotted: horizontal retrace.

COLOUR DEPTH

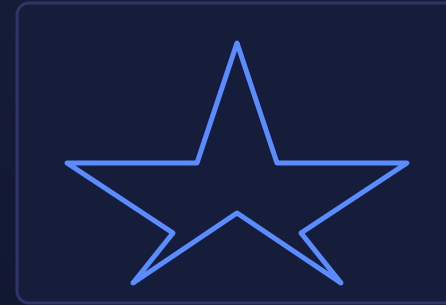
Bits per pixel decide the number of colours: 8 bits gives 256 colours, 24 bits gives true colour (about 16.7 million).

Random Scan (Vector) Displays

The beam draws only the lines that make up the picture

Also called **vector**, **stroke writing** or **calligraphic** displays. The picture is stored as a set of line drawing commands in a **display list** (refresh display file), and the beam traces those lines directly.

- Refresh rate depends on the **number of lines** to draw.
- Produces very smooth, high quality **line drawings**.
- Cannot easily show shaded, filled or photographic images.



The beam follows the shape outline, not a pixel grid.

Raster vs Random Scan

Two very different ways to build a picture

Raster Scan

- Picture is a matrix of pixels
- Stored in a frame buffer
- Great for shaded, realistic scenes
- Lines can look jagged (staircase effect)
- Used in almost all modern displays

vs

Random Scan

- Picture is a set of line segments
- Stored in a display list
- Great for smooth line art
- Cannot show filled or photo images well
- Mostly obsolete today

Colour CRT Monitors

Producing colour by combining light from phosphors

Beam Penetration

- Used with **random scan** monitors.
- Two phosphor layers: red and green.
- Beam speed and voltage decide how deep it penetrates.
- Only **four colours**: red, green, orange, yellow.

Shadow Mask

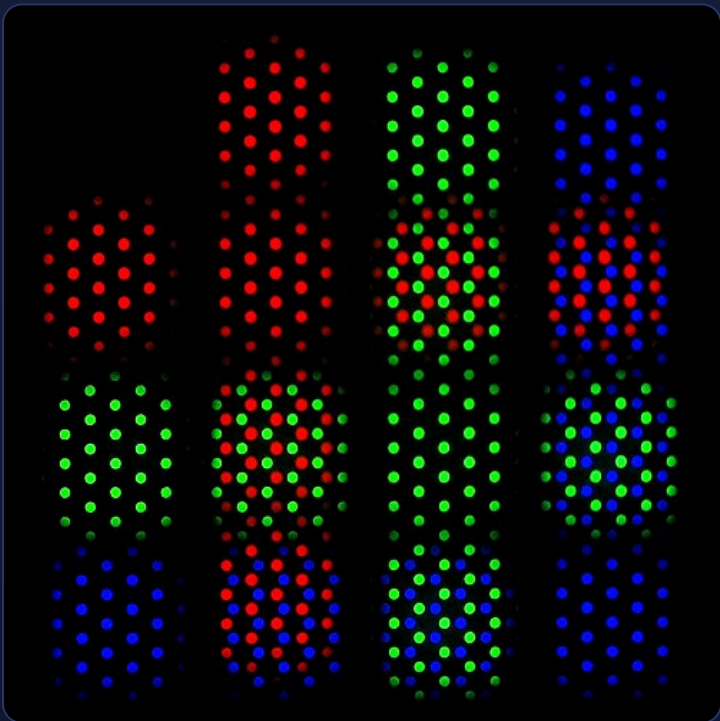
- Used with **raster scan**; produces millions of colours.
- Each pixel is a triad of **red, green and blue** phosphor dots.
- Three electron guns, one per colour.
- A perforated **shadow mask** ensures each beam hits only its own dot.



Overlapping R, G, B light
combines into any colour (RGB r

Full red plus green plus blue gives white; varying intensities give every colour.

REAL PHOTO · 40X MAGNIFICATION



Shadow mask of a 15 inch CRT under about 40x magnification, showing the RGB phosphor matrix. Source: Selcuk Oral (Drumex), Wikimedia Commons, CC BY-SA 3.0.

Flat Panel Displays

Thinner, lighter and lower power than the CRT

Emissive

Convert electrical energy directly into light.



Plasma panels

Gas discharge between glass plates.



Electroluminescent

Thin film that glows under voltage.



LED

Light emitting diodes.

Non Emissive

Use optical effects to turn ambient light or a backlight into a picture.

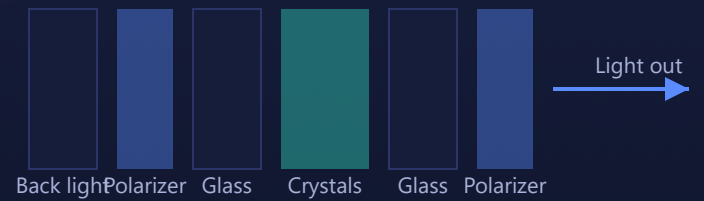


LCD

Liquid Crystal Displays, the dominant non emissive type.

Liquid Crystal Displays (LCD)

Light is passed or blocked by twisting crystals



Voltage twists the crystals to pass or block polarized light.

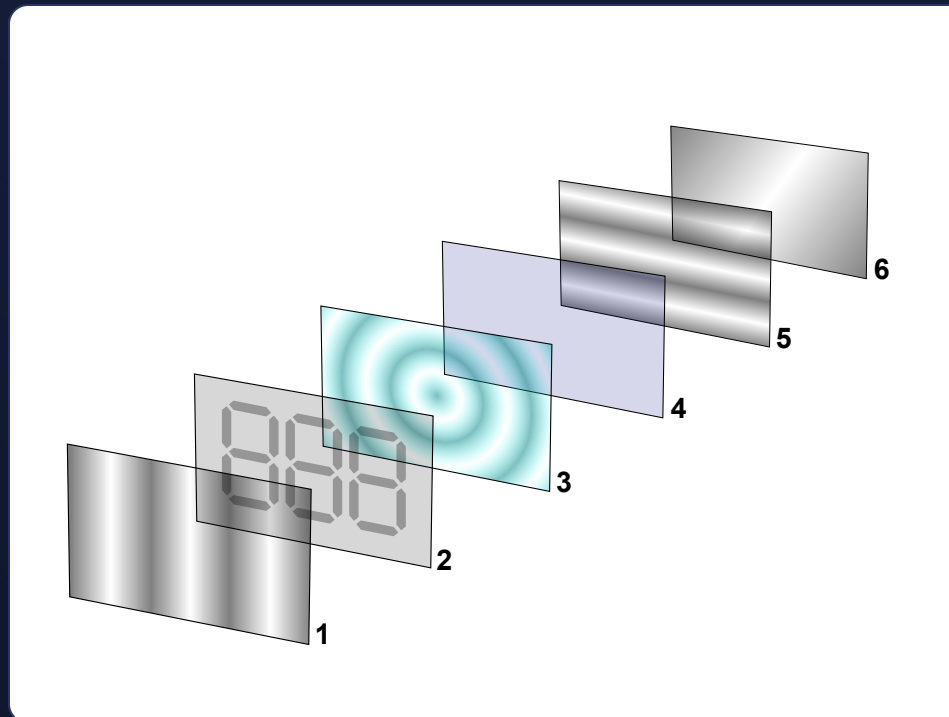
Passive vs Active matrix

- **Passive matrix:** a grid of conductors addresses pixels by row and column.
- **Active matrix (TFT):** a thin film transistor at every pixel gives sharper, faster images with better contrast.

NON EMISSIVE

An LCD makes no light of its own, so it needs a backlight, today usually LEDs.

REFERENCE IMAGE



Layers of a reflective twisted nematic LCD: polarizers, glass with ITO electrodes, twisted crystals, reflector. Source: Ed g2s, Wikimedia Commons, CC BY-SA 3.0.

Graphics Workstations and Viewing Systems

Complete systems for professional graphics work



High resolution

Large screens, often 2560 x 2048 or higher, sometimes multiple monitors.



Fast GPUs

Powerful graphics processors for real time rendering.



Stereoscopic

Separate left and right eye views create a sense of depth.



Virtual reality

Head mounted displays with head tracking immerse the user in a scene.

USED IN

CAD engineering, animation studios, scientific visualization, simulation and training environments.

Computer Graphics Software

From coordinate systems and graphics functions to the OpenGL API, GLUT window management, a complete program, and error handling.

Graphics Software Categories

General programming packages

A library of graphics functions callable from a language such as C, C++ or Java. They provide primitives, attributes, transformations, viewing and input.

Example: OpenGL

Special purpose applications

Ready made packages designed for non programmers to create pictures or models in a specific domain.

Examples: CAD, paint and draw apps, medical imaging

STANDARDS MAKE SOFTWARE PORTABLE

Older standards were GKS and PHIGS. Today **OpenGL** is the cross platform standard that runs the same code on many systems.

Coordinate Representations

A scene passes through several coordinate systems before it reaches the screen



Objects move from their own local space to the pixels on screen.

- **Modelling (local) coordinates:** each object is defined in its own frame.
- **World coordinates:** all objects are placed in one common scene.
- **Normalized coordinates:** mapped to a standard 0 to 1 range for portability.
- **Device (screen) coordinates:** final integer pixel positions on the display.

Graphics Functions

A general package groups its functions into these categories



Primitives

Points, lines, polygons and text, the basic building blocks.



Attributes

Colour, line style, fill pattern and text font.



Transformations

Translate, rotate and scale objects.



Viewing

Choose what part of the scene is shown and where.



Input

Handle mouse, keyboard and other devices.



Control

Initialize, manage errors and talk to the system.

Introduction to OpenGL

A cross platform, hardware independent graphics API

- **OpenGL** means Open Graphics Library, a device independent API.
- Provides a few hundred functions for 2D and 3D graphics.
- Works as a **state machine**: you set states such as colour, then draw.
- Functions begin with `gl`; constants begin with `GL_`.

Naming convention

A suffix tells you the **argument count** and **data type**.

```
glColor3f(...) // 3 args, float  
glVertex2i(...) // 2 args, integer  
glVertex3fv(...) // 3 float, vector
```

Types: `i` integer, `f` float, `d` double, `v` array.

Related Libraries

OpenGL is usually paired with helper libraries

Library	Prefix	Purpose
OpenGL core (GL)	gl	Core drawing and state functions
OpenGL Utility (GLU)	glu	Higher level routines: quadrics, curves, viewing setup
OpenGL Utility Toolkit (GLUT)	glut	Windowing, menus and input, platform independent
System glue (GLX, WGL)	varies	Connect OpenGL to the operating system window

WHY GLUT?

Core OpenGL knows nothing about windows or input. GLUT creates the display window, reads the mouse and keyboard, and runs the event loop in a portable way.

Display Window Management with GLUT

The standard sequence to open a window and begin drawing

```
glutInit(&argc, argv);           // initialize GLUT
glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB); // buffer and colour mode
glutInitWindowPosition(50, 100); // top left corner
glutInitWindowSize(400, 300);   // width, height
glutCreateWindow("My First OpenGL Program"); // title bar text
```

- `GLUT_SINGLE` uses one frame buffer; `GLUT_DOUBLE` uses double buffering for smooth animation.
- `GLUT_RGB` selects RGB colour mode.
- `glutDisplayFunc(fn)` registers the callback that draws the scene.
- `glutMainLoop()` starts the event loop and does not return.

A Complete OpenGL Program

Draw a line segment, the classic first program

```
#include <GL/glut.h>

void init(void) {
    glClearColor(1.0, 1.0, 1.0, 0.0); // white background
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(0.0, 200.0, 0.0, 150.0); // 2D world window
}

void lineSegment(void) {
    glClear(GL_COLOR_BUFFER_BIT); // clear the window
    glColor3f(0.0, 0.0, 1.0); // blue
    glBegin(GL_LINES);
        glVertex2i(180, 15);
        glVertex2i(10, 145);
    glEnd();
    glFlush(); // send to display
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowPosition(50, 100);
    glutInitWindowSize(400, 300);
    glutCreateWindow("Line Segment");
    init();
    glutDisplayFunc(lineSegment);
    glutMainLoop();
    return 0;
}
```

Error Handling in OpenGL

OpenGL records errors in a flag you can query at any time

When a function is called with invalid arguments or in an invalid state, an **error flag** is set. You read it with `glGetError()` and turn the code into text with `gluErrorString()`.

```
GLenum code = glGetError();
if (code != GL_NO_ERROR)
    printf("%s", gluErrorString(code));
```

Common error codes

Code	Meaning
GL_NO_ERROR	No error
GL_INVALID_ENUM	Bad enum argument
GL_INVALID_VALUE	Numeric argument out of range
GL_INVALID_OPERATION	Operation not allowed in this state
GL_OUT_OF_MEMORY	Not enough memory

Module 1 Summary

Hardware

- CRT: electron beam plus phosphor, needs refresh.
- Raster uses a pixel matrix and frame buffer; random uses a line list.
- Colour CRT: beam penetration and shadow mask (RGB).
- Flat panels: emissive (plasma, LED) and non emissive (LCD).

Software

- Coordinate pipeline: modelling to world to device.
- Function groups: primitives, attributes, transforms, viewing, input.
- OpenGL plus GLU plus GLUT, using a state machine model.
- Complete program structure and `glGetError`.

NEXT UP

Module 2, Graphics Output Primitives and their Attributes, Chapters 3 and 4.

Sources & References

Textbook chapters, official documentation and image credits for this module

Primary reading

- Hearn, Baker & Carithers, **Computer Graphics with OpenGL**, 4th ed., Pearson, 2014, **Chapter 1** (A Survey of Computer Graphics) and **Chapter 2** (Overview of Graphics Systems).
- Official API docs: Khronos OpenGL Registry, OpenGL / GLU reference pages.

Image credits

- CRT diagram: Interiot / Theresa Knott, [Wikimedia Commons](#), CC BY-SA 3.0.
- Shadow mask close-up: Selcuk Oral (Drumex), [Wikimedia Commons](#), CC BY-SA 3.0.
- LCD layers: Ed g2s, [Wikimedia Commons](#), CC BY-SA 3.0.

LECTURER

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